

Lab 1 — Non-linear Regression & Model Selection

Unit 1: Classification & Non-linear Regression

J. Balsa • Machine Learning II • Academic year 2026–2027

Context

This laboratory applies Unit 1 concepts to a pharmacokinetic dataset. We will model the plasma concentration of an orally administered drug over time using sigmoidal basis functions, and select the model by cross-validation.

1. Starting material

A bioavailability study followed a group of volunteers with serial blood draws over 24 hours after a single oral dose. Two files are provided: `pk_train.csv` (70 samples) and `pk_test.csv` (35 samples), with columns `t_h`, the time since the dose in hours, and `C_mgL`, the measured plasma concentration in mg/L.

```
1 import pandas as pd
2
3 train = pd.read_csv("pk_train.csv")
4 test = pd.read_csv("pk_test.csv")
5
6 X_train = train[["t_h"]].values
7 y_train = train["C_mgL"].values
```

The true form of the curve is unknown to you. You will approximate it with a model that is linear in the parameters and non-linear in t , using sigmoidal basis functions

$$\phi_j(t) = \frac{1}{1 + \exp\left(-\frac{t - \mu_j}{\text{width}}\right)},$$

with `n_centers` centres μ_j equally spaced over the 24-hour study window.

Hint

Use the provided Jupyter Notebook skeleton. Fill in the missing code blocks to complete the tasks. Do not touch the test file until the last exercise.

2. Tasks

Exercise 1 — Basis Functions

We build the design matrix and look at what the basis actually does.

Exercise 1(a) — Data Exploration

Load both files and plot the concentration against time.

Exercise 1(b) — The Sigmoidal Design Matrix

Write a function `sigmoid_basis(X, n_centers, width)` that maps a column of times to the design matrix Φ , with one column per basis function.

Exercise 1(c) — What does each hyperparameter do?

Plot the basis functions for several values of `n_centers` and `width`. Which one plays the role of the polynomial degree?

Exercise 2 — Model Selection

Training error cannot choose hyperparameters, since a more flexible model always fits the training data better. We use cross-validation instead.

Exercise 2(a) — Grid Search with Cross-Validation

Build a pipeline that applies the basis and then Ridge regression, and search over `n_centers`, `width` and the regularization parameter λ with 5-fold cross-validation. Report the winning combination and its CV error, and work out how many models were fitted in total.

Exercise 2(b) — The U-shaped Curve

Plot the cross-validation error against λ on a logarithmic axis, holding `n_centers` and `width` at their best values. Explain each arm of the U in terms of bias and variance, and inspect the fitted weights at both ends of the range.

Exercise 3 — Generalization

Finally we estimate out-of-sample performance.

Exercise 3(a) — Test Set Evaluation

Score the test set once. Report the test RMSE and R^2 , and plot the fitted curve over both sets. Compare the cross-validation error with the test error: which is larger, and why was that to be expected?